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Production and Financial Networks in Interplay

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1 Big picture

- **Question.** How do bank credit supply shocks propagate through the production network of firms, and how much do these network spillovers amplify aggregate output losses?
- **Core idea.** Financial and production networks are not separate. A bank shock to one firm changes that firm's demand for inputs and cost of supplying outputs, which affects its suppliers and customers, and then propagates further through the supply chain.
- **Setting.** The paper studies Spain during the global financial crisis, using administrative data on the universe of firm-to-firm VAT transactions and bank-firm loans.
- **Identification.** Bank credit supply shocks are identified from cross-bank variation in lending during the crisis, using the Amiti-Weinstein / Khwaja-Mian approach with firm fixed effects in bank-firm credit growth regressions.
- **Main reduced-form fact.** A one standard deviation negative bank shock to a customer's bank supply reduces a supplier's sales to that customer by about 2.36 percentage points. A similar shock to a supplier reduces its sales to customers by about 1.09 percentage points.
- **Direction asymmetry.** Upstream propagation, from customers to suppliers, is stronger than downstream propagation, from suppliers to customers.
- **Node-level result.** At the firm level, bank shocks to direct customers reduce firm sales and employment. The direct effect of a firm's own bank shock is smaller than the customer-shock propagation effect.
- **Structural contribution.** The paper builds a general equilibrium production-network model with financial frictions modeled as input-price wedges, then structurally estimates link-level, node-level, and aggregate propagation effects.
- **Higher-order spillovers.** Shocks do not stop at direct suppliers or customers. The structural model shows that higher-order network effects are comparable in magnitude to first-order effects.
- **Aggregate result.** Bank shocks reduce Spanish GDP by roughly 2.35 to 3.94 percent in the model with the production network. Without input-output propagation, the fall is about 1.78 to 2.25 percent.
- **Policy takeaway.** Because bank shocks spread through the production network, financial distress in banks can have real effects far beyond the firms directly borrowing from those banks.

2 Data and measurement

2.1 Administrative production-network data

- The production network is constructed from Spain's confidential VAT register.
- Firms report annual transactions with third parties exceeding 3,005 euros.
- The data contain both the supplier-reported and customer-reported value of each transaction.
- The authors use annual firm-to-firm transactions in 2008 and 2009 to form the weighted production network.
- The raw network contains more than 13.8 million transactions among 867,013 firms in 2008 and nearly 12.0 million transactions among 861,350 firms in 2009.
- Each directed link ji represents annual sales from supplier j to customer i .

Interpretation. The VAT register lets the authors observe the actual supplier-customer network rather than relying on industry-level input-output tables. This is crucial because the paper studies propagation at the firm-link level.

2.2 Bank-firm credit data

- Bank lending relationships come from Spain’s credit register, maintained by the Bank of Spain.
- The register includes loan commitments above 6,000 euros granted by banks operating in Spain.
- Loan-level information is aggregated to the bank-firm-year level.
- The analysis focuses on depository financial institutions lending to nonfinancial firms.
- Because Spain is bank dominated, bank credit shocks are especially relevant for firm financing and real activity.

Interpretation. The bank-firm credit register allows the authors to map every borrowing firm to the banks that supplied its credit before the crisis. This makes it possible to construct firm-level exposure to bank credit supply shocks.

2.3 Firm and bank balance-sheet data

- Firm balance sheets and income statements come from the Spanish mercantile register.
- These data provide controls such as firm size, age, liquidity, leverage, maturity structure, legal form, sales, and employment.
- Bank balance-sheet data provide information about bank size and net interbank borrowing.
- Interbank borrowing is important because banks with higher precrisis interbank funding exposure were more affected by the global financial crisis.

Interpretation. The combination of firm controls, bank controls, and network data allows the empirical design to separate bank supply shocks from firm demand and local industry shocks.

2.4 Bank shock construction

The baseline bank shock follows the Amiti-Weinstein approach. For bank b lending to firm i , estimate:

$$CreditGrowth_{ib} = \nu_i + \iota_b + \epsilon_{ib}. \quad (1)$$

- $CreditGrowth_{ib}$ is lending growth from bank b to firm i over 2008.
- ν_i is a firm fixed effect, absorbing firm credit demand and other firm fundamentals.
- ι_b is the bank fixed effect, interpreted as a bank-level credit supply shock.
- The firm-level bank shock is a precrisis-credit-weighted average of bank fixed effects across the firm’s lenders.
- The sign is reversed so that higher values mean a more negative bank credit supply shock.

Interpretation. A firm that borrowed before the crisis from banks that later contracted lending more sharply receives a larger negative bank shock. The firm fixed effects in the first-stage bank-firm regression are what makes this a credit supply measure rather than a credit demand measure.

2.5 Network shock definitions

- A **direct bank shock** is the shock hitting the focal firm’s own lenders.
- An **upstream shock** is a shock hitting a customer’s lenders and propagating upstream to the supplier.
- A **downstream shock** is a shock hitting a supplier’s lenders and propagating downstream to the customer.
- A **first-order shock** comes from direct suppliers or direct customers.
- A **higher-order shock** comes from indirect suppliers/customers along longer paths in the production network.

Interpretation. The paper’s key innovation is to measure and model both direct and indirect exposure to bank shocks, including shocks that travel along long network paths.

3 Empirical specifications and models

3.1 Reduced-form link-level regression

Let s_{ji} be sales from supplier j to customer i . The link-level regression is

$$\Delta \log s_{ji} = a_u \theta_i^{AW} + a_d \theta_j^{AW} + x_{ji} + \epsilon_{ji}. \quad (2)$$

- θ_i^{AW} is the customer bank shock.
- θ_j^{AW} is the supplier bank shock.
- a_u measures upstream propagation from customers to suppliers.
- a_d measures downstream propagation from suppliers to customers.
- x_{ji} includes supplier, customer, link, industry, location, and fixed-effect controls depending on the specification.

Interpretation. The link-level design compares different supplier-customer links while absorbing many forms of supplier, customer, industry, and geography heterogeneity. It asks whether a link declines more when the customer or supplier is exposed to a bank shock.

3.2 Reduced-form node-level regression

At the firm level, the authors aggregate shocks from all direct customers and all direct suppliers. The generic node-level regression is

$$\Delta \log y_i = a_0 \theta_i^{AW} + a_u \Theta_i^C + a_d \Theta_i^S + x_i + \epsilon_i, \quad (3)$$

where Θ_i^C is the sales-share-weighted shock to firm i ’s customers and Θ_i^S is the input-share-weighted shock to its suppliers.

Interpretation. The node-level regressions ask whether a firm’s total sales or employment decline because its customers or suppliers are exposed to constrained banks.

3.3 Production-network model

Each firm produces a single good with Cobb-Douglas inputs and a CES composite of intermediate inputs:

$$y_i = \zeta_i k_i^\rho \ell_i^\beta M_i^\alpha, \quad (4)$$

$$M_i = \left(\sum_{j \in N} g_{ji}^{1/\sigma} z_{ji}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}. \quad (5)$$

- α , β , and ρ are input shares of intermediate inputs, labor, and capital.
- σ is the elasticity of substitution across intermediate inputs.
- $G = (g_{ji})$ is the input-output network matrix.

Interpretation. The parameter σ is central. If customers can substitute across suppliers, downstream shocks are weakened. The estimates imply $\sigma > 1$, but not high enough to eliminate propagation.

3.4 Financial friction as an input-price wedge

A firm must finance a share χ_i of input expenditure in advance. If the bank interest rate is R_i , the effective wedge is

$$\theta_i = \chi_i R_i. \quad (6)$$

Its profit can be written as

$$\pi_i = p_i y_i - (1 + \theta_i) \left(\sum_{j \in N_i^+} p_j z_{ji} + w \ell_i + r k_i \right). \quad (7)$$

Interpretation. Bank shocks operate as cost wedges on all inputs. A constrained bank raises the effective cost of producing for the borrowing firm.

3.5 Structural link-level regression

The empirical counterpart of the structural link equation is

$$\Delta \log \left(\frac{s_{ji}}{s_i} \right) = \lambda_F^d \theta_j^{AW} + \lambda_F^u \theta_i^{AW} + \lambda_H^d Net_j^{AW} + \lambda_H^u Net_i^{AW} + x_{ji} + \varepsilon_{ji}. \quad (8)$$

- λ_F^u and λ_F^d are first-order upstream and downstream effects.
- λ_H^u and λ_H^d are higher-order upstream and downstream effects.
- The structural mapping implies the elasticity of substitution can be recovered as

$$\sigma = 1 + \lambda_F^d / \lambda_F^u.$$

Interpretation. The structural link equation adds higher-order network terms to the reduced-form link regression. This lets the paper measure what the reduced form misses.

3.6 Structural node-level equation

Define the matrix $H = \alpha V^{-1} G M V$, whose elements capture the importance of customer i for supplier j . The node-level structural equation is

$$d \log \left(\frac{s_i}{E} \right) = -e_i' (I - H)^{-1} H \theta + (1 - \sigma) e_i' \Lambda \theta. \quad (9)$$

- The first term is an upstream demand channel.
- The second term is a bidirectional channel combining downstream cost propagation and upstream demand propagation.

Interpretation. A bank shock to one firm changes costs, prices, input demand, and downstream substitution patterns. Those adjustments then propagate upstream as demand shocks.

3.7 Aggregate GDP equation

The model implies a closed-form approximation to the effect of bank shocks on real GDP, represented by aggregate consumption c :

$$d \log c = \text{intermediate-input channel} + \text{extwagechannel} + \text{extcapital} - \text{returnchannel}. \quad (10)$$

In the paper's notation:

$$d \log c = -\gamma'[I - \alpha G']^{-1} \theta -$$

$$\frac{\beta}{1 - \alpha} d \log w -$$

$$\frac{\rho}{1 - \alpha} d \log r. \quad (11)$$

Interpretation. The aggregate effect is not just a sum of firm-level shocks. It includes the whole production network, wage adjustment, capital return adjustment, and both first- and higher-order propagation.

4 Identification and empirical design

4.1 Why the bank shocks are credible

- The bank shock is estimated from bank-firm lending growth with firm fixed effects, so it is designed to remove firm demand for credit.
- Firm-level shocks are weighted averages of precrisis bank relationships, not relationships chosen after the crisis.
- Firms exposed to more constrained banks look similar to other firms on precrisis observable characteristics.
- The shocks matter for firm credit availability during the crisis, but not before the crisis.
- Banks with more precrisis interbank borrowing and smaller size are more likely to contract credit during the crisis, matching the crisis narrative.

Interpretation. The identifying variation comes from firms being attached before the crisis to banks that were hit differently by the global financial crisis.

4.2 Why link-level evidence is powerful

- Link-level regressions observe over one million supplier-customer links.
- Supplier fixed effects can compare sales from the same supplier to different customers.
- Customer fixed effects can compare purchases by the same customer from different suppliers.
- Link-level controls capture relationship importance, same bank, same location, and same industry/geography factors.

Interpretation. If the same supplier sells less to customers hit by negative bank shocks, the result is difficult to explain by a supplier-specific downturn alone.

4.3 Why the structural model is needed

- Reduced-form link regressions mainly capture first-order direct-neighbor effects.
- A production network allows shocks to travel through suppliers' suppliers, customers' customers, and many longer paths.

- General equilibrium effects change prices, wages, capital returns, and substitution incentives.
- The structural model provides a coherent way to aggregate all paths into link-level, node-level, and GDP effects.

Interpretation. Without the model, the empirical exercise would understate total propagation because it would omit higher-order and equilibrium channels.

4.4 Robustness and alternative explanations

- The paper uses interbank borrowing exposure as an instrument for bank shocks.
- It also instruments reductions in bank debt with the bank shock.
- It checks robustness to alternative ways of constructing bank shocks and to concerns about bank specialization and loan type.
- It considers whether local supplier-customer relationships and local banks jointly drive the results.
- It shows stronger propagation effects for more financially constrained firms, such as smaller or more leveraged firms.

Interpretation. The robustness checks aim to show that the results come from credit supply shocks and production-network propagation, not from common local or sectoral demand shocks.

4.5 Main limitations

- The analysis is based on Spain during the global financial crisis, a bank-dominated economy; external validity to other countries depends on financial structure.
- VAT transactions identify domestic supplier-customer relationships, but exports and some transactions outside the VAT system are less central in the network.
- The structural model imposes parametric production, substitution, and markup assumptions.
- Higher-order effects are model-based, so their interpretation depends on the validity of the network model.
- The analysis studies the short-run 2008-2009 crisis window, not long-run rewiring of networks.

Interpretation. The paper provides unusually granular evidence, but its most aggregate claims necessarily combine data with a structural model.

5 Tables, equation panels, and figures

The paper has no standalone graphical figures. The visual material below directly crops the main original tables and selected original equation panels from the paper, grouped to save space.

5.1 Model panels: production, financial wedge, node equation, and GDP equation

The production side of the economy consists of a given set of firms, N , each of them producing a single good with a technology displaying constant returns to scale. The production of a typical firm i is described by the production function:³⁰

$$(4) \quad y_i = f_i(k_i, \ell_i, M_i) = \zeta_i k_i^\rho \ell_i^\beta M_i^\alpha,$$

where y_i stands for the output of firm i , k_i for the physical capital used, ℓ_i for its labor input, M_i is the following CES aggregate of the intermediate inputs:

$$(5) \quad M_i = \left(\sum_{j \in N} g_{ji}^{\frac{1}{\sigma}} z_{ji}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}},$$

where z_{ji} stands for the amount of intermediate input j used by firm i . Thus, $f_i(\cdot)$ displays the nested CES form, with the strictly positive α , β , and ρ being input shares,³¹ σ is the elasticity of substitution across intermediate inputs, and $\zeta = (\zeta_i)_{i=1}^n$ is the vector of Hicks productivity parameters. The nonnegative vector $(g_{ji})_{j \in N_i^+}$ reflects the *relative* intensity with which any given firm i uses different intermediate inputs in the set N_i^+ and satisfies $\sum_{j \in N} g_{ji} = 1$ for every i . The interfirm (technological) production structure of the economy is then characterized by the (column-stochastic) adjacency matrix $\mathbf{G} = (g_{ji})_{ji=1}^n$.

Figure 1: Production function and CES input aggregator

Read:

- The production function combines capital, labor, and a CES aggregate of intermediate inputs.
- The CES aggregator makes the elasticity of substitution across intermediate inputs, σ , central for shock propagation.
- Bank shocks are represented by $\theta_i = \chi_i R_i$, a wedge that raises the effective cost of inputs.
- Higher bank financing costs therefore operate as production-cost shocks, not merely as balance-sheet shocks.

Economic message. The model turns bank distress into a firm-level cost wedge that then propagates through supplier-customer relationships.

We first note that the matrix $\mathbf{H} \equiv \alpha \mathbf{V}^{-1} \mathbf{G} \mathbf{M} \mathbf{V}$ has elements $h_{ji} = \frac{s_{ji}}{s_j}$ that essentially capture the relative importance of firm i among all direct customers of j (see Lemma 3 in Supplemental Appendix B). This information is observed in our dataset and hence can be directly used in our empirical analysis. Therefore, we write Λ in the following more convenient form:

$$\Lambda = (\mathbf{I} - \mathbf{H})^{-1} (\text{diag}(\mathbf{H}\mathbf{I}) - \mathbf{H}\mathbf{G}') (\mathbf{I} - \alpha \mathbf{G}')^{-1},$$

which then allows us to rewrite (9) as follows:

$$(10) \quad \overbrace{\frac{d \log \left(\frac{s_i}{E} \right)}{\text{Change of Domar weight of firm } i}} = \overbrace{-\mathbf{e}_i' (\mathbf{I} - \mathbf{H})^{-1} \mathbf{H} \mathbf{0}}^{\text{Upstream shock}} + (1 - \sigma) \overbrace{\mathbf{e}_i' \Lambda \mathbf{0}}^{\text{Bidirectional shock}}.$$

Figure 3: Node-level upstream and bidirectional channels

As in Bigio and La'O (2020), we assume that every firm i is required to pay in advance a share χ_i of its input expenditure, which is financed by borrowing at an interest rate R_i . Its net profit is then given by:

$$\begin{aligned} \pi_i &= p_i y_i - (1 - \chi_i) \left(\sum_{j \in N_i^+} p_j z_{ji} + w \ell_i + r k_i \right) \\ &\quad - \chi_i (1 + R_i) \left(\sum_{j \in N_i^+} p_j z_{ji} + w \ell_i + r k_i \right) \\ &= p_i y_i - (1 + \theta_i) \left(\sum_{j \in N_i^+} p_j z_{ji} + w \ell_i + r k_i \right), \end{aligned}$$

Figure 2: Bank-shock input-financing wedge

$$(8R) \quad \Delta \log \left(\frac{s_{ji}}{s_i} \right) = \lambda_j^L \theta_j^{AW} + \lambda_i^L \theta_i^{AW} + \lambda_{ji}^L \text{Net}_j^{AW} + \lambda_{ji}^L \text{Net}_i^{AW} + \mathbf{x}_{ji} + \varepsilon_{ji},$$

Figure 4: Structural empirical link equation (8R)

$$(11) \quad d\log c = -\gamma' [\mathbf{I} - \alpha \mathbf{G}]^{-1} \boldsymbol{\theta} - \frac{\beta}{1 - \alpha} d\log w - \frac{\rho}{1 - \alpha} d\log r,$$

where

$$(12) \quad d\log w = \frac{\eta}{1 + \eta} \frac{\beta}{wL} \mathbf{s}' \mathbf{M} d\log \mathbf{s} - \frac{1 - \delta}{1 + \eta} d\log c,$$

$$(13) \quad d\log r = \frac{\rho}{rK} \mathbf{s}' \mathbf{M} d\log \mathbf{s},$$

and

$$(14) \quad d\log \mathbf{s} = -[\mathbf{I} - \mathbf{H}]^{-1} \mathbf{H} \boldsymbol{\theta} + (1 - \sigma) \boldsymbol{\Lambda} \boldsymbol{\theta}.$$

Figure 5: Aggregate GDP equations

Read:

- Equation (10) decomposes node-level sales effects into an upstream shock and a bidirectional shock.
- The upstream term captures a demand effect: constrained firms demand fewer inputs from suppliers.
- The bidirectional term captures downstream cost effects, price changes, substitution, and subsequent upstream demand effects.
- Equation (8R) is the empirical link-level structural regression with first-order and higher-order shocks.
- The GDP equations aggregate direct shocks, input-output propagation, wage adjustment, and capital-return adjustment.

Economic message. These panels show how the paper moves from micro link-level effects to node-level propagation and aggregate macro effects.

5.2 Table 1 and Table 2: reduced-form link-level upstream and downstream propagation

TABLE 1—LINK-LEVEL: PROPAGATION OF BANK SHOCKS THROUGH THE NETWORK OF CUSTOMERS: REDUCED FORM, UPSTREAM PROPAGATION (INDIRECT SHOCKS VIA BANK SHOCKS TO FIRST-ORDER CUSTOMERS)

Dependent variable: $\Delta \log$ (sales from supplier to customer)			IV. Instrument: Bank net interbank borrowing		IV. Instrument: Bank shock	
	(1)	(2)	First stage (3)	2 ^o stage (4)	First stage (5)	2 ^o stage (6)
Direct (bank) shock	-0.703 (0.415)					
Customer (bank) shock (first order effect)		-2.359 (1.181)		-6.068 (2.628)	3.445 (0.571)	
Customer (bank) net interbank borrowing			8.917 (0.933)			
Customer reduction of bank debt						-0.739 (0.265)
Customer:						
Controls	—	Yes	Yes	Yes	Yes	Yes
Spatial $\times$ Industry fixed effects	—	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	No	No	No	No	No
Firm:						
Controls	Yes	—	—	—	—	—
Spatial $\times$ Industry fixed effects	Yes	—	—	—	—	—
Fixed effects	No	Yes	Yes	Yes	Yes	Yes
Firm $\times$ Supplier spatial and industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
First stage effective F -statistic	—	—	65.28	—	36.09	—
R^2	0.404	0.474		—		—
Observations	1,119,169	1,119,169	1,119,169	1,119,169	1,119,169	1,119,169

Table 1: Table 1: link-level upstream propagation from customer bank shocks

TABLE 2—LINK-LEVEL: PROPAGATION OF BANK SHOCKS THROUGH THE NETWORK OF SUPPLIERS: REDUCED FORM, DOWNSTREAM PROPAGATION (INDIRECT SHOCKS VIA BANK SHOCKS TO FIRST-ORDER SUPPLIERS)

Dependent variable: $\Delta \log$ (sales from supplier to customer)	IV. Instrument: Bank net interbank borrowing					
			IV. Instrument: Bank shock			
	(1)	(2)	First stage (3)	2 ^o stage (4)	First stage (5)	2 ^o stage (6)
Direct (bank) shock	-2.678 (1.050)					
Supplier (bank) shock (first order effect)		-1.086 (0.546)		-3.656 (1.249)	2.094 (0.499)	
Supplier (bank) net interbank borrowing			9.227 (0.623)			
Supplier reduction of bank debt						-0.519 (0.310)
Supplier:						
Controls	—	Yes	Yes	Yes	Yes	Yes
Spatial $\times$ Industry fixed effects	—	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	No	No	No	No	No
Firm:						
Controls	Yes	—	—	—	—	—
Spatial $\times$ Industry fixed effects	Yes	—	—	—	—	—
Fixed effects	No	Yes	Yes	Yes	Yes	Yes
Firm $\times$ Supplier spatial and industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
First stage effective F -statistic	—	—	97.53	—	12.40	—
R^2	0.358	0.483	—	—	—	—
Observations	1,114,420	1,114,420	1,114,420	1,114,420	1,114,420	1,114,420

Table 2: Table 2: link-level downstream propagation from supplier bank shocks

Read:

- Table 1 estimates upstream propagation. A one standard deviation customer bank shock reduces supplier-customer sales growth by 2.359 percentage points in the controlled specification.
- The interbank-borrowing IV estimate is larger in magnitude: -6.068 percentage points.
- When customer bank debt reduction is instrumented by the bank shock, the second-stage coefficient is -0.739.
- Table 2 estimates downstream propagation. A one standard deviation supplier bank shock reduces sales to customers by 1.086 percentage points.
- The downstream IV estimate is -3.656, and the bank-debt reduction IV coefficient is -0.519.
- Both tables have more than 1.1 million link-level observations.

Economic message. A customer's credit shock causes a stronger reduction in supplier sales than a supplier's credit shock causes in customer purchases. The directional asymmetry is one of the paper's central empirical facts.

5.3 Table 3: reduced-form node-level sales and employment effects

TABLE 3—NODE-LEVEL: FIRM-LEVEL EFFECTS OF BANK SHOCKS THROUGH THE PRODUCTION NETWORK: REDUCED FORM, FIRM-LEVEL SALES AND EMPLOYMENT

Dependent variable:	$\Delta \log(\text{sales})$		$\Delta \log(\text{employment})$	
	(1)	(2)	(3)	(4)
Direct (bank) shock	-0.888 (0.520)	-0.821 (0.490)	-0.552 (0.241)	-0.536 (0.239)
Customer (bank) shock (first order effect)		-2.223 (0.373)		-0.488 (0.091)
Supplier (bank) shock (first order effect)		0.374 (0.346)		-0.031 (0.069)
Firm controls	Yes	Yes	Yes	Yes
Spatial and industry fixed effects	Yes	Yes	Yes	Yes
R^2	0.351	0.353	0.091	0.091
Observations	196,171	196,171	196,171	196,171

Notes: This table reports estimates from WLS for columns 1 and 2 and OLS for the rest of the columns. See Section IV. Observations are at the level of the firm (node-level). The dependent variables are the change, between 2008 and 2009, in the log of aggregate sales to all customers (columns 1 and 2) and the log of total employment (columns 3 and 4). Bank shock

Table 3: Table 3: node-level reduced-form effects on sales and employment

Read:

- Direct bank shocks reduce firm sales: the coefficient is -0.888 in column 1 and -0.821 in column 2.
- Customer bank shocks also reduce firm sales: -2.223 in column 2.
- Supplier bank shocks do not significantly reduce firm sales in the node-level reduced form; the coefficient is 0.374.
- For employment, direct bank shocks have negative coefficients around -0.55.
- Customer bank shocks reduce employment by -0.488, while supplier shocks have little effect.

Economic message. At the firm level, upstream propagation through customers shows up in both sales and employment. Downstream propagation is weaker once firm-level aggregation and substitution are taken into account.

5.4 Table 4: structural link-level propagation

TABLE 4—LINK-LEVEL: PROPAGATION OF BANK SHOCKS THROUGH THE PRODUCTION NETWORK:
STRUCTURAL FORM

Dependent variable: $\Delta \log$ (sales from supplier to customer/sales of customer)	Upstream propagation		Downstream propagation		Joint estimation nonstandardized	Nonstandardized
	(1)	(2)	(3)	(4)	(5)	(6)
Customer (bank) shock (first order effect)	-1.924 (0.900)	-2.002 (0.924)			-42.411 (20.242)	-35.510 (16.535)
Higher order customer (bank) shock		1.796 (0.632)			51.176 (18.520)	45.724 (19.771)
Supplier (bank) shock (first order effect)			-1.086 (0.546)	-1.055 (0.527)	-23.950 (12.621)	-15.149 (8.543)
Higher order supplier (bank) shock				-2.045 (1.066)	-47.837 (28.554)	-55.222 (32.630)
Supplier/Customer: Controls	Yes	Yes	Yes	Yes	Yes	Yes
Spatial $\times$ Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm: Controls	—	—	—	—	—	Yes
Spatial $\times$ Industry fixed effects	—	—	—	—	—	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	No
Fixed effects as supplier or customer	No	No	No	No	No	Yes
Firm $\times$ Supplier/Customer spatial and industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.477	0.478	0.493	0.493	0.485	0.449
Observations	1,119,169	1,119,169	1,114,420	1,114,420	2,233,589	1,313,776

Table 4: Table 4: link-level structural estimates

Read:

- The structural first-order upstream effect is about -1.924 to -2.002, close to the reduced-form Table 1 effect.
- The first-order downstream effect is about -1.086 to -1.055, close to the reduced-form Table 2 effect.
- Higher-order customer shocks have a positive coefficient of 1.796 in column 2.
- Higher-order supplier shocks have a negative coefficient of -2.045 in column 4.
- In the joint nonstandardized specification, the first-order customer effect is -42.411 and the first-order supplier effect is -23.950.
- The nonstandardized estimates imply an elasticity of substitution across intermediate inputs of about $\sigma = 1.56$.

Economic message. Reduced-form regressions capture only part of propagation. The structural estimates show that higher-order network effects are large and that upstream effects exceed downstream effects because substitution attenuates downstream cost propagation.

5.5 Table 5: structural node-level propagation

TABLE 5—NODE-LEVEL: PROPAGATION OF BANK SHOCKS THROUGH THE PRODUCTION NETWORK:
STRUCTURAL FORM

Dependent variable: $\Delta \log(\text{sales})$	(1)	(2)	(3)	(4)	Nonstandardized (5)
Upstream (first and higher order effects)	-1.810 (0.326)		-1.611 (0.333)	-1.641 (0.350)	-32.891 (6.802)
Bidirectional (up and down first and higher order effects)		-1.435 (0.333)	-0.527 (0.300)	-0.439 (0.367)	-11.441 (6.504)
Direct (bank) shock				-0.411 (0.537)	
Firm controls	Yes	Yes	Yes	Yes	Yes
Spatial and industry fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.358	0.357	0.358	0.358	0.358
Observations	196,171	196,171	196,171	196,171	196,171

Notes: This table reports estimates from WLS. See Section VI. Observations are at the level of the firm (node-level).

Table 5: Table 5: node-level structural estimates

Read:

- The upstream channel is negative and significant: -1.810 in column 1.
- The bidirectional channel is also negative and significant: -1.435 in column 2.
- When both channels are included, upstream is -1.611 and bidirectional is -0.527.
- Adding the direct bank shock in column 4 leaves upstream and bidirectional propagation negative, while the direct bank shock becomes statistically insignificant.
- Nonstandardized coefficients in column 5 imply an alternative estimate of $\sigma = 1.35$, close to the link-level estimate of 1.56.

Economic message. Once the model aggregates all network shocks, the direct bank shock loses much of its standalone explanatory power. This supports the model’s view that firm outcomes are shaped by the equilibrium propagation of all shocks in the economy.

6 Short summary

- The paper studies how bank credit shocks propagate through the production network of Spanish firms during the global financial crisis.
- It combines VAT-based supplier-customer transaction data, credit-register bank-firm loan data, firm balance sheets, and bank balance sheets.
- Bank shocks are constructed using the Amiti-Weinstein approach and precrisis bank relationships.
- Link-level reduced-form estimates show negative upstream and downstream propagation.
- Upstream propagation is stronger: customer bank shocks reduce supplier-customer sales by 2.36 percentage points, while supplier bank shocks reduce customer purchases by 1.09 percentage points.
- Node-level estimates show that customer shocks reduce firm sales and employment; supplier shocks are weaker at the node level.
- The general equilibrium model represents bank shocks as input-financing wedges and production as a firm-level network with CES intermediate inputs.
- Structural estimates recover an elasticity of substitution across intermediate inputs of about 1.56 at the link level and 1.35 at the node level.

- Higher-order network effects are economically important and comparable to first-order effects.
- Bank shocks reduce GDP by roughly 2.35 to 3.94 percent when network propagation is included.
- Without input-output propagation, the implied GDP decline is only about 1.78 to 2.25 percent.
- Production-network propagation therefore amplifies the aggregate impact of bank shocks by close to 50 percent.
- First-order network effects account for about 53 percent of total network amplification; higher-order effects account for the rest.

7 Conclusion

7.1 Core conclusion

Huremovic, Jimenez, Moral-Benito, Peydro, and Vega-Redondo show that bank shocks to firms propagate through production networks and that this propagation materially amplifies the real effects of financial crises. The effect is not confined to direct borrowers of constrained banks. It travels through suppliers, customers, and longer supply-chain paths.

7.2 Economic intuition

A customer hit by a bank credit shock buys fewer inputs, hurting its suppliers. A supplier hit by a bank credit shock becomes more expensive or less able to supply, hurting customers, but customers can partly substitute across suppliers. This substitution channel explains why upstream propagation is stronger than downstream propagation. At the aggregate level, many such micro spillovers interact through prices, wages, capital returns, and the production network, making the crisis larger than the direct bank shocks alone.

7.3 Takeaway

The paper’s central lesson is that financial shocks should be studied jointly with the real production network. A bank shock to one firm can become a real shock to many firms that never borrowed from the affected bank. Higher-order links matter: distant suppliers and customers are not negligible. For macro-financial policy, stabilizing banks can have large real effects because it prevents shocks from spreading through supply chains.